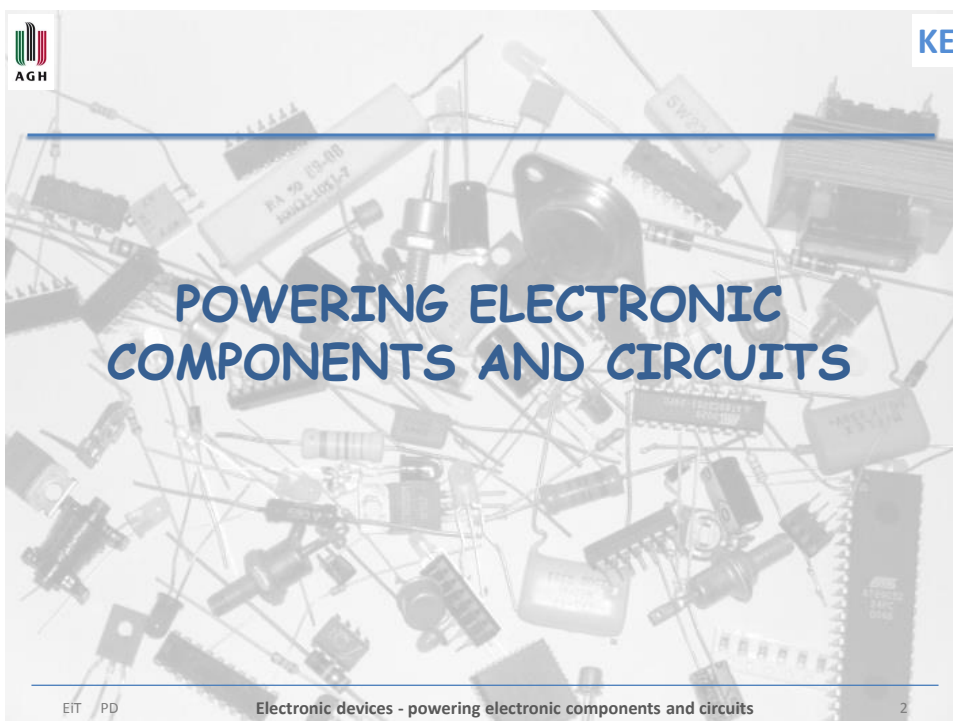




 AGH UNIVERSITY OF SCIENCE AND TECHNOLOGY
IM. STANISŁAWA STASZICA W KRAKOWIE
Faculty of Computer Science, Electronics and Telecommunications
DEPARTMENT OF ELECTRONICS

ELECTRONIC DEVICES

Piotr Dziurdzia, Ph.D.
C-3, room 413; tel. 617-27-02, Piotr.Dziurdzia@agh.edu.pl



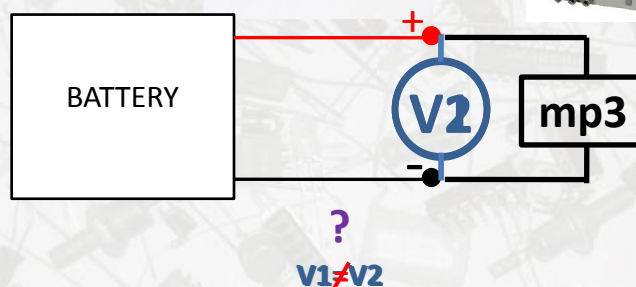
 

POWERING ELECTRONIC COMPONENTS AND CIRCUITS

EIT PD Electronic devices - powering electronic components and circuits 2

POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

VOLTAGE SOURCES



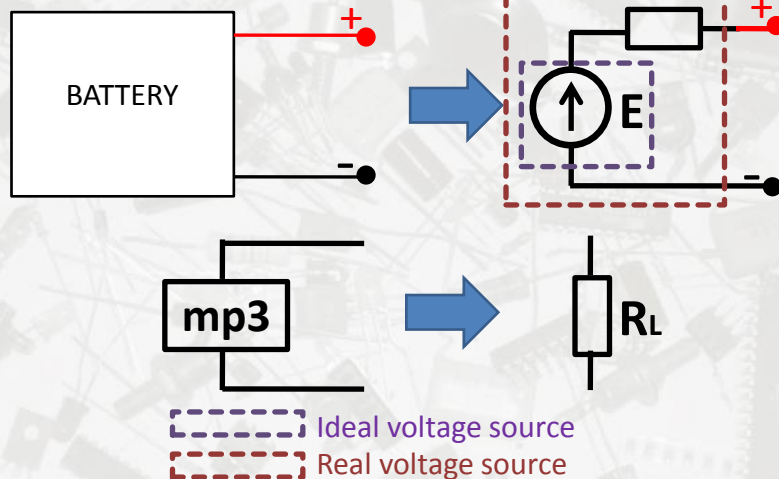
EIT PD

Electronic devices - powering electronic components and circuits

3

POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

VOLTAGE SOURCES



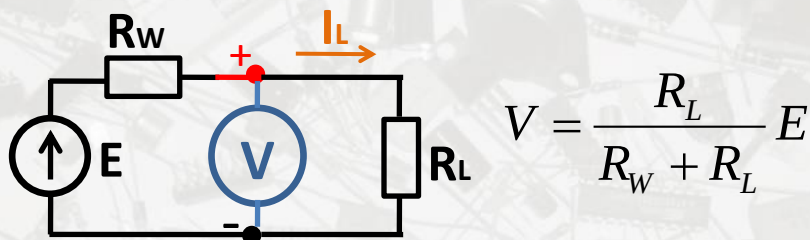
EIT PD

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4

POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

VOLTAGE SOURCES



EIT PD

Electronic devices - powering electronic components and circuits

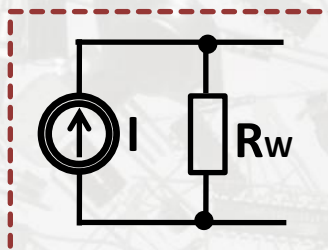
5

POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

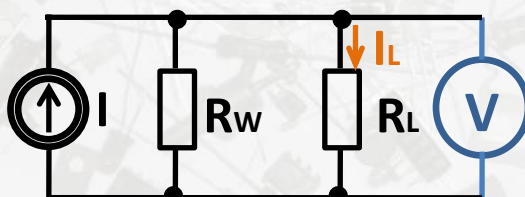
CURRENT SOURCES



Ideal current source



Real current source



$$I_L = \frac{R_W}{R_W + R_L} I$$

$$V = \frac{R_W \cdot R_L}{R_W + R_L} I$$

EIT PD

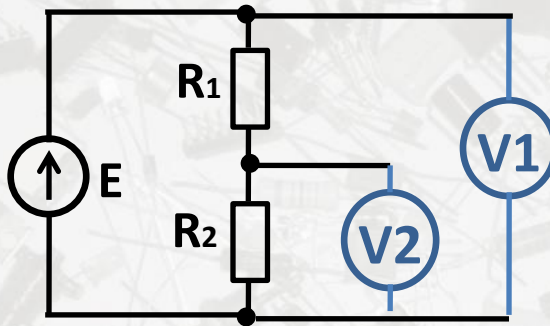
Electronic devices - powering electronic components and circuits

6



POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

VOLTAGE DIVIDER



$$V1 = E$$

$$V2 = \frac{R_2}{R_1 + R_2} E$$

if $E=9V$, then:

$$R1=R2 \rightarrow V2=4.5V$$

$$R1=2R2 \rightarrow V2=3.0V$$

$$R2=2R1 \rightarrow V2=6.0V$$

EIT PD

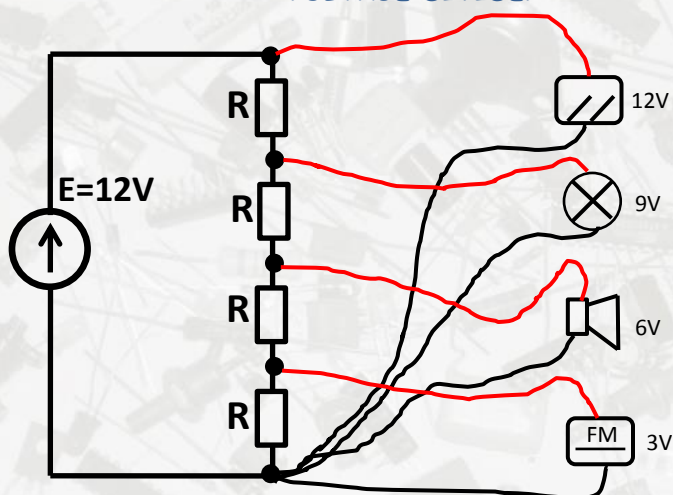
Electronic devices - powering electronic components and circuits

7



POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

VOLTAGE DIVIDER



EIT PD

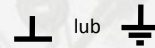
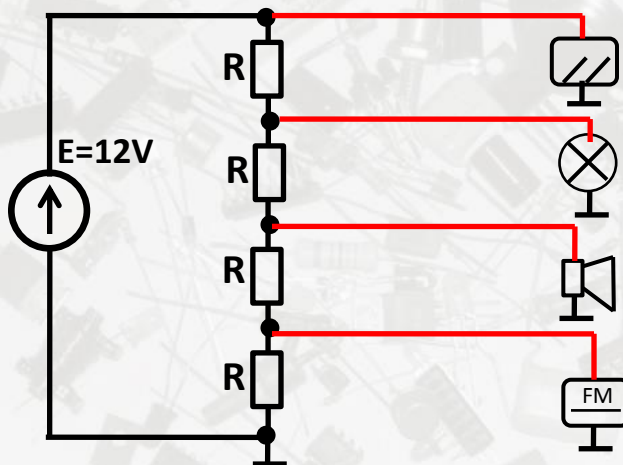
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8



POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

VOLTAGE DIVIDER



„GROUND” of
electronic circuit

Common potential of the
circuit – it facilitates
design of the schematic
diagrams and real
electronic circuits.

EIT PD

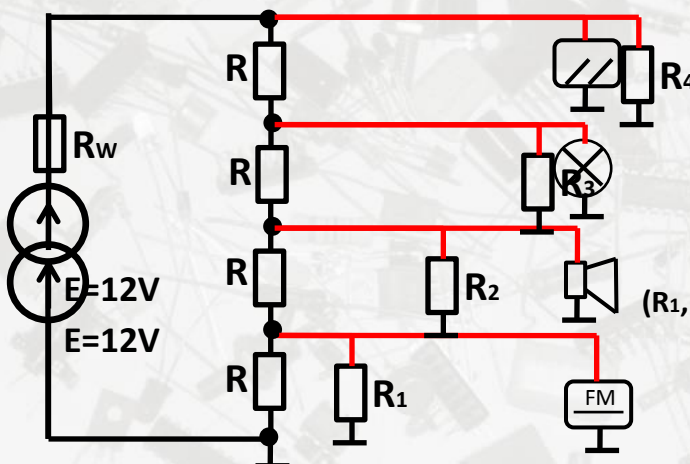
Electronic devices - powering electronic components and circuits

9



POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

VOLTAGE DIVIDER - ANNEX WHEN CAN WE FACE A PROBLEM?



Voltage divider
„works” properly
if:

$$R_w \ll 4R$$

and

$$(R_1, R_2, R_3, R_4) \gg R$$

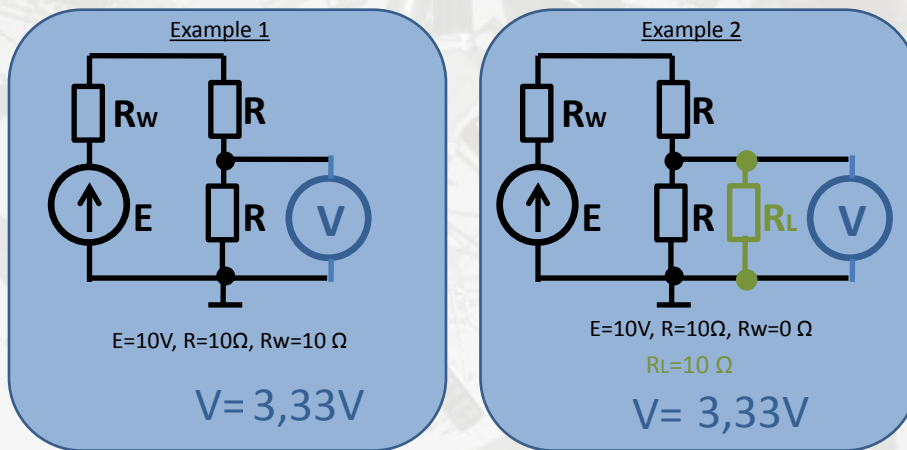
EIT PD

Electronic devices - powering electronic components and circuits

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POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

VOLTAGE DIVIDER - ANNEX WHEN CAN WE FACE A PROBLEM?



EIT PD

Electronic devices - powering electronic components and circuits

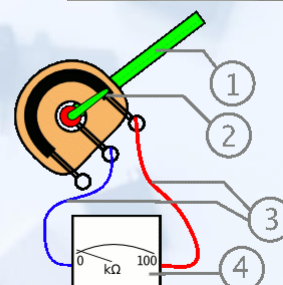
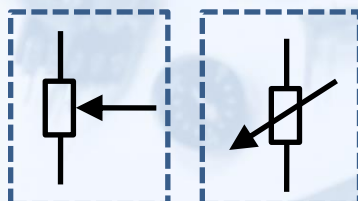
11

POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

POTENTIOMETER - A VOLTAGE DIVIDER PROVIDING SMOOTH ADJUSTMENT OF VOLTAGE RATIO



Resistive track is made of carbon, cermet, plastic or a coil of wire



Source: Created with The GIMP, Piotr Jaworski

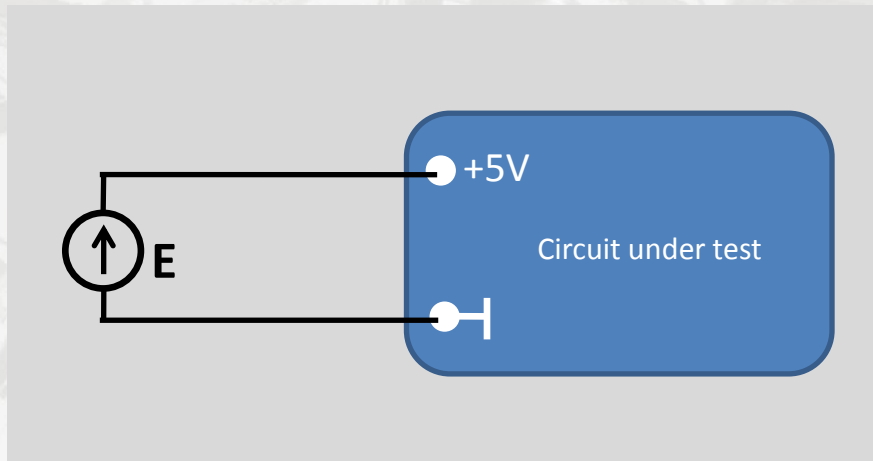
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POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

SUPPLYING ELECTRONIC CIRCUITS



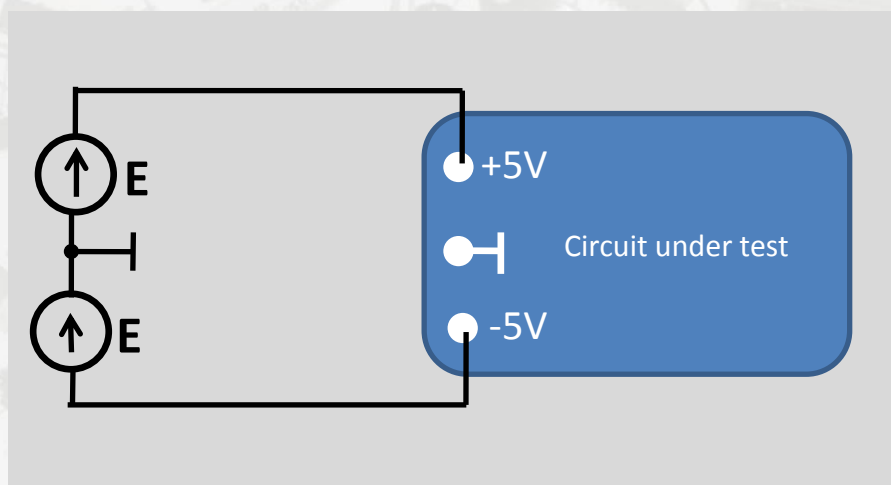
EIT PD

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POWERING ELECTRONIC COMPONENTS AND CIRCUITS ^{KE}

SUPPLYING ELECTRONIC CIRCUITS

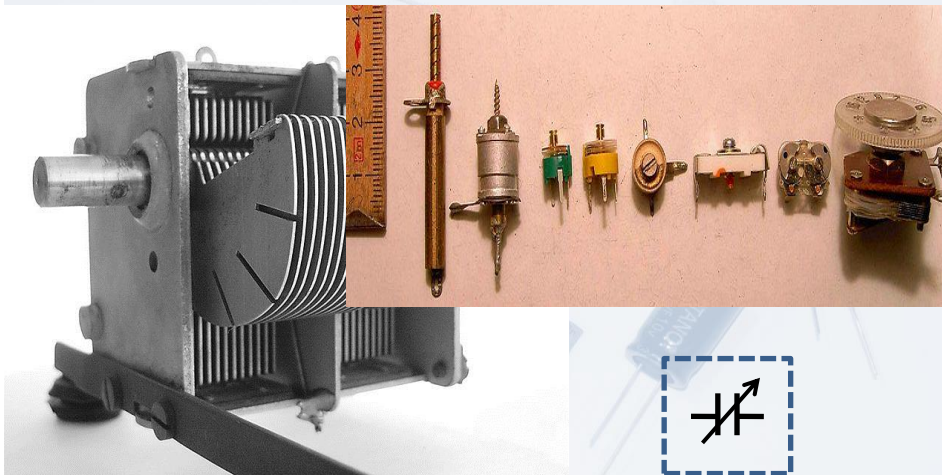


EIT PD

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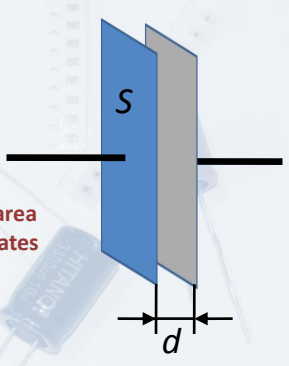
?? ? CAPACITOR ? ? ?



CAPACITOR

For flat (plate) capacitor:

permittivity of the vacuum ϵ_0
 relative permittivity of the dielectric ϵ_r
 surface area of the plates S
 capacitor capacitance C
 distance between plates d

$$C = \frac{\epsilon_0 \epsilon_r S}{d}$$


CAPACITOR



capacitors: ceramic, folic, paper

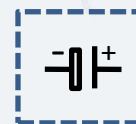


Electrolytic capacitor



PARAMETERS:

- nominal voltage U_N
- losses $\tan \delta$
- tolerance %
- value of capacitance C

units: μF , nF , pF 

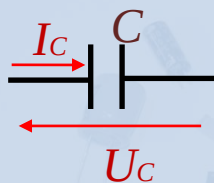
EIT PD

Electronic devices - capacitor

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CAPACITOR

Ideal capacitor



$$C = \frac{Q}{U}$$

$$I = \frac{dQ}{dt}$$

$$I_C = C \frac{dU_C}{dt}$$

TO REMEMBER

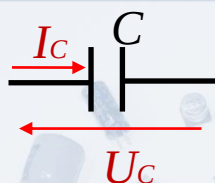
for R electrical current is proportional to the voltage
for C current is proportional to the rate of change of the voltage

EIT PD

Electronic devices - capacitor

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CAPACITOR



Ideal capacitor

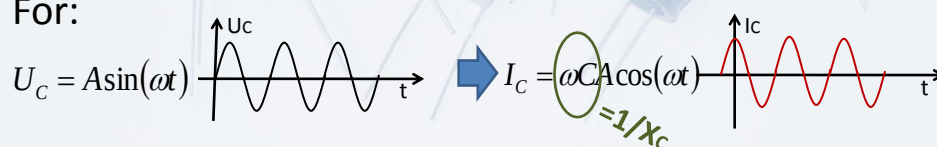
If:

$$I_C = C \frac{dU_C}{dt} \quad \text{then}$$

For:

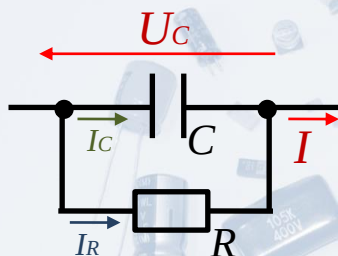


For:



CAPACITOR

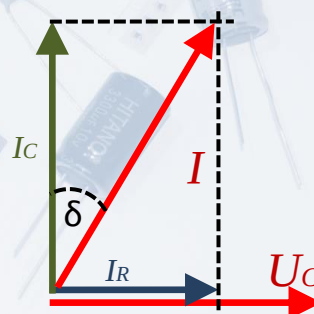
Non-ideal capacitor



For alternating sinusoidal signals, the voltages and currents can be presented in form of vectors

Losses in the capacitor

$$\tan \delta = \frac{I_R}{I_C} = \frac{\frac{U}{R}}{U \omega C} = \frac{1}{\omega R C}$$

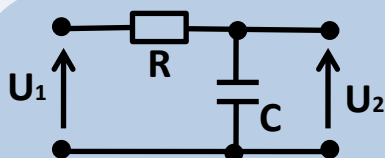


CAPACITOR

KE

RC IN ELECTRONIC CIRCUITS

INTEGRATOR

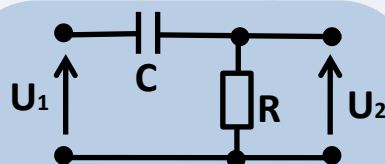


$$\tau = RC$$

$$u_1 = \tau \frac{du_2}{dt} + u_2$$

$$U_2(s) = U_1(s) \frac{1}{1 + s\tau}$$

DIFFERENTIATOR



$$\tau = RC$$

$$\tau \frac{du_2}{dt} + u_2 = \tau \frac{du_1}{dt}$$

$$U_2(s) = U_1(s) \frac{s\tau}{1 + s\tau}$$

EIT PD

Electronic devices - capacitor

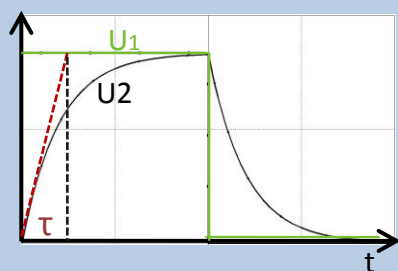
21

CAPACITOR

KE

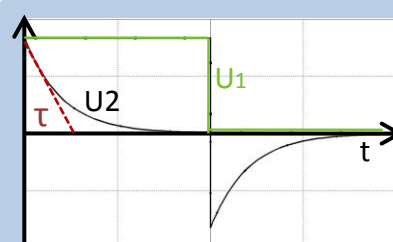
RESPONSE OF THE RC CIRCUITS TO A SQUARE SIGNAL - TRANSIENT ANALYSIS

INTEGRATOR



$$U_2(t) = U_1 \left(1 - e^{-\frac{t}{\tau}} \right)$$

DIFFERENTIATOR



$$U_2(t) = U_1 e^{-\frac{t}{\tau}}$$

EIT PD

Electronic devices - capacitor

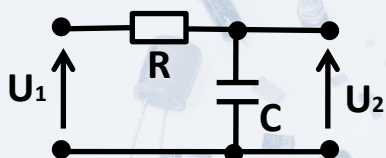
22

CAPACITOR

KE

RESPONSE OF THE RC CIRCUITS TO A SQUARE SIGNAL - STEADY STATE

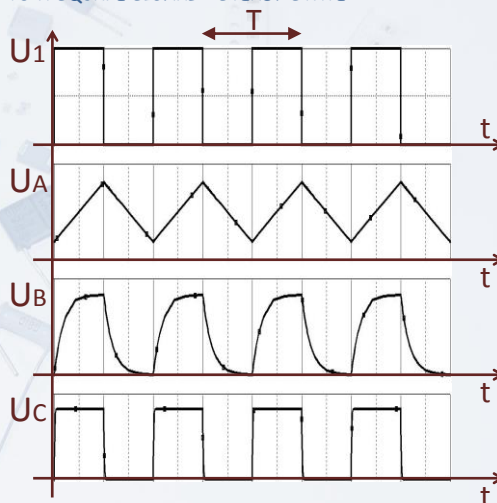
INTEGRATOR



$U_2 = U_B ? \Rightarrow$ yes, if $\tau \approx T$

$U_2 = U_A ? \Rightarrow$ yes, if $\tau \gg T$

$U_2 = U_C ? \Rightarrow$ yes, if $\tau \ll T$



EIT PD

Electronic devices - capacitor

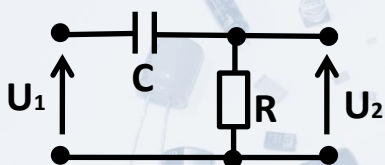
23

CAPACITOR

KE

RESPONSE OF THE RC CIRCUITS TO A SQUARE SIGNAL - STEADY STATE

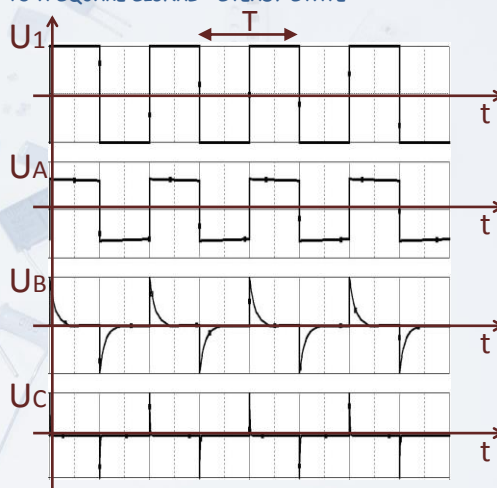
DIFFERENTIATOR



$U_2 = U_B ? \Rightarrow$ yes, if $\tau \approx T$

$U_2 = U_A ? \Rightarrow$ yes, if $\tau \gg T$

$U_2 = U_C ? \Rightarrow$ yes, if $\tau \ll T$



EIT PD

Electronic devices - capacitor

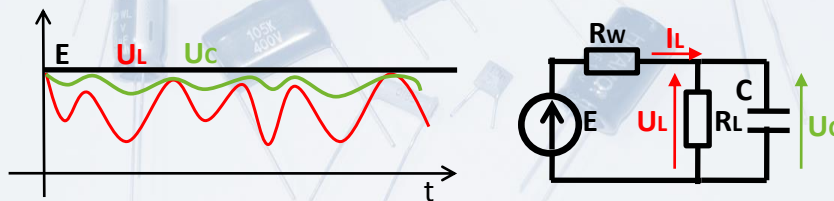
24

CAPACITOR

KE

CAPACITORS - APPLICATIONS

With the ability to store energy, capacitors exhibit inertia and allow for maintaining the instantaneous voltage values in systems where there is a pulse current consumption (power circuits, against interfere, etc.)



EIT PD

Electronic devices - capacitor

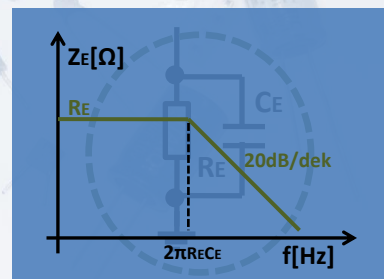
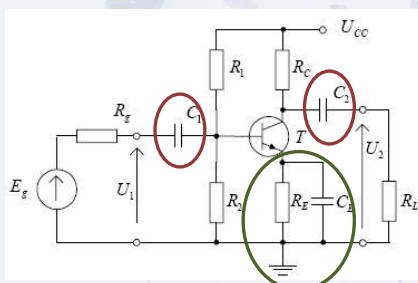
25

CAPACITOR

KE

CAPACITORS - APPLICATIONS

COUPLING CAPACITORS - they block the penetration of DC signals between the signal source (such as a generator) and an amplifier, as well as between receiver (eg. speaker) and the amplifier.



$$Z_E = f(f) = ?$$

EIT PD

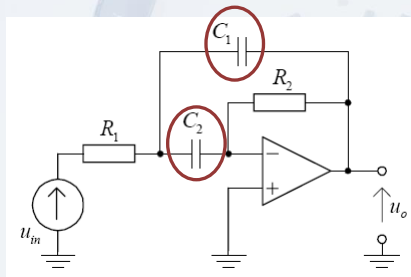
Electronic devices - capacitor

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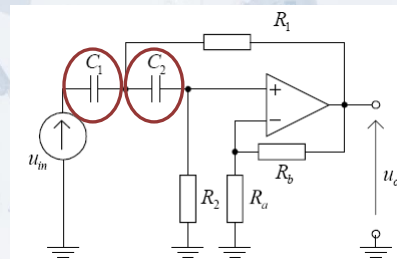
CAPACITOR

CAPACITORS - APPLICATIONS

FILTER CAPACITORS - essential elements in shaping the frequency characteristics of electronic systems (including amplifiers)



Band-pass filter



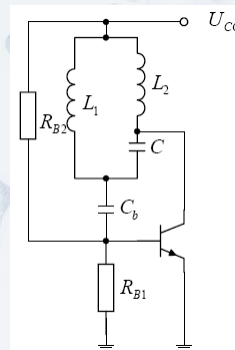
High-pass filter

CAPACITOR

CAPACITORS - APPLICATIONS

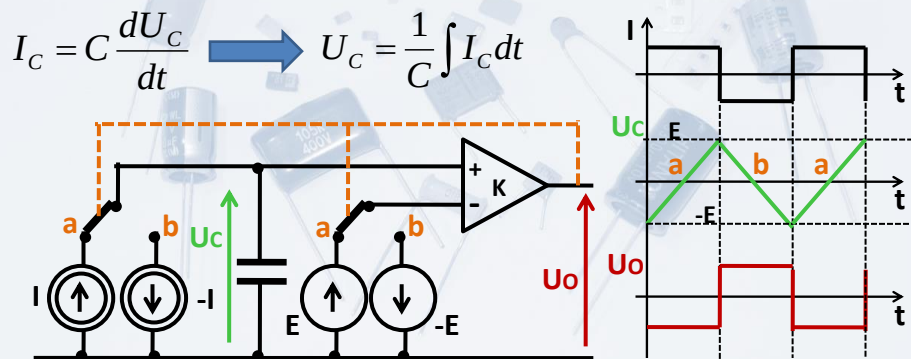
CAPACITORS in sinusoidal voltage generators

$$f = \frac{1}{2\pi\sqrt{LC}}$$



CAPACITOR

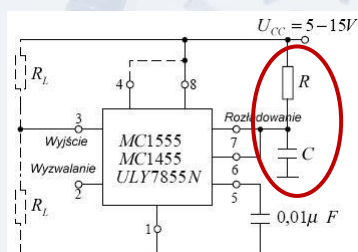
CAPACITORS in generators - time dependency between voltage and current in the capacitors is used to determine the time relationships in square and saw-tooth waveform generators



CAPACITOR

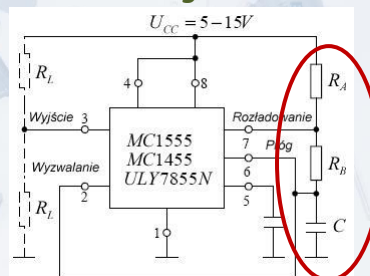
CAPACITORS in timing circuits - for example, the '555 timer flip-flop circuits (monostable and astable generators)

Monostable generator



$$f_m = \frac{0,9}{RC}$$

Astable generator

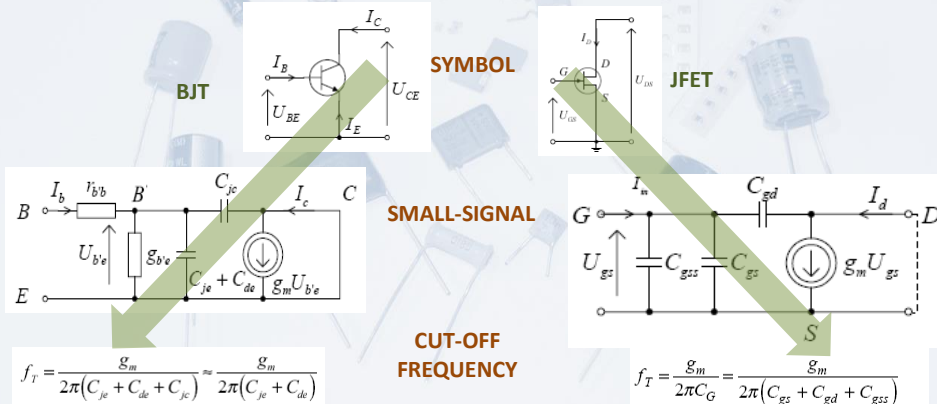


$$f_a = \frac{1,44}{(R_A + 2R_B)C}$$

CAPACITOR

WHERE DO CAPACITORS (CAPACITANCES) CREATE OBSTACLES?

Capacitances between the electrodes of transistors limit the maximum operating frequency



EIT PD

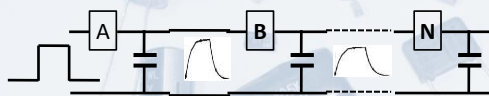
Electronic devices - capacitor

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CAPACITOR

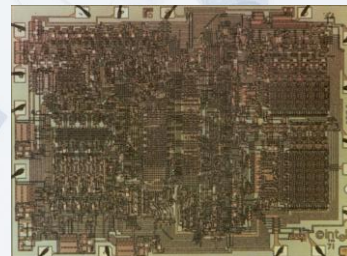
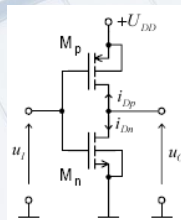
WHERE DO CAPACITORS (CAPACITANCES) CREATE OBSTACLES?

Parasitic capacitances of signal paths and the inputs gates cause the increasing propagation time



In CMOS circuits

$$P = fCU^2$$



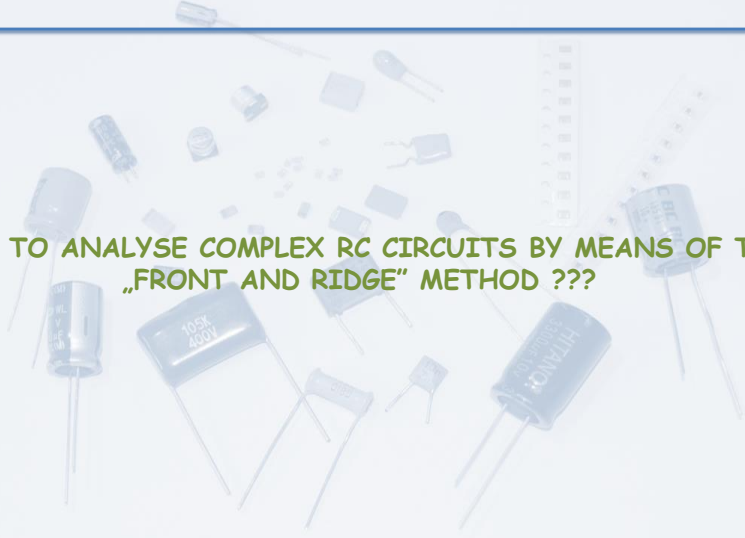
EIT PD

Electronic devices - capacitor

32

KE

CAPACITOR



HOW TO ANALYSE COMPLEX RC CIRCUITS BY MEANS OF THE „FRONT AND RIDGE” METHOD ???

EIT PD

Electronic devices - capacitor

33

KE

Two pieces of wire



Do they play the same role in the electronic circuit?

EIT PD

Electronic devices

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INDUCTIVE COIL is a:

two-terminal passive (conservative) element
capable of collecting energy in form of a
magnetic field.
It is described by inductance L [H].

EIT PD Electronic devices - inductive coil 36



INDUCTIVITY

KE

$$L = \frac{\psi}{i}$$

L – inductivity,
 Ψ - magnetic flux associated with the coil,
 i - current flowing in the coil

$$u = \frac{d\psi}{dt}$$

u – voltage at the coil terminals,
 t – time

$$u = L \frac{di}{dt}$$

Voltage is proportional to the rate of change of the current

EIT PD

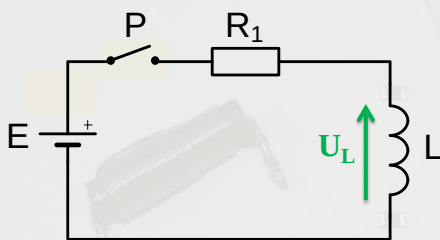
Electronic devices - inductive coil

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COIL IN ELECTRICAL CIRCUIT

KE



$$I = \frac{E}{R_1}$$

$$U_L = L \frac{dI}{dt} = 0$$

in a steady state

In a real system, voltage rises to such an extent that it can cause a breakdown, leading to a destruction of a device

$$U_L = -\infty$$

for an ideal coil

EIT PD

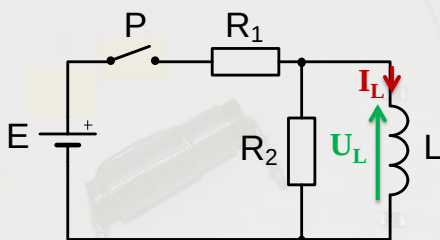
Electronic devices - inductive coil

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COIL IN ELECTRICAL CIRCUIT

KE



In this arrangement, the voltage U_L rises to the $-E$, and the current will decrease exponentially – energy in the coil will be "discharged" by the R_2

$$I = \frac{E}{R_1}$$

$$U_L = L \frac{dI}{dt} = 0$$

in a steady state

$$U_L = -E e^{-\frac{t}{\tau}}$$

$$I_L = E \frac{R_2}{R_1} e^{-\frac{t}{\tau}}$$

time constant: $\tau = \frac{L}{R_2}$

EIT PD

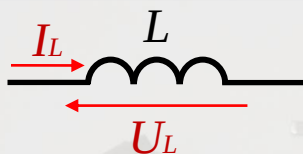
Electronic devices - inductive coil

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AN IDEAL COIL

KE



If:

$$U_L = L \frac{dI_L}{dt}$$

then

For:



For:



EIT PD

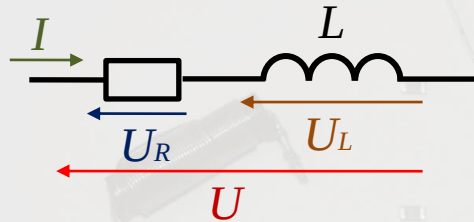
Electronic devices - inductive coil

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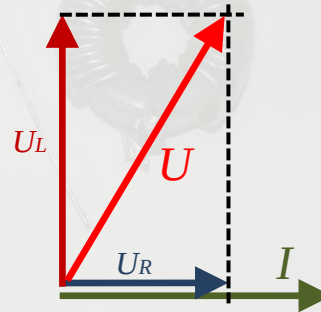


A REAL COIL

KE



For alternating sinusoidal signals, the voltages and currents can be presented in form of vectors



Quality factor

$$Q = \frac{|U_L|}{U_R} = \frac{|I\omega L|}{IR} = \frac{\omega L}{R}$$

EIT PD

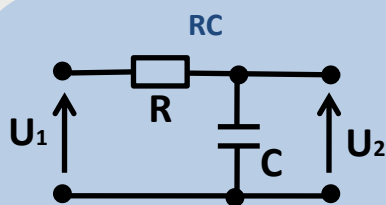
Electronic devices – inductive coil

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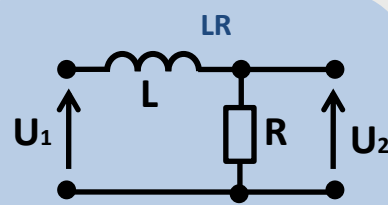
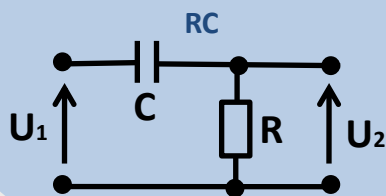


INTEGRATOR AND DIFFERENTIATOR

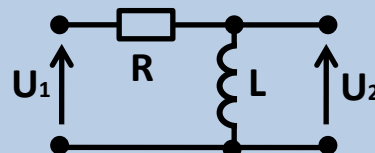
KE



$$\tau = RC$$



$$\tau = \frac{L}{R}$$



EIT PD

Electronic devices - inductive coil

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A take home message

for R electrical current is proportional to the voltage

$$I_R = U_R \frac{1}{R}$$

for C current is proportional to the rate of change of the voltage

$$I_C = C \frac{dU_C}{dt}$$

for L voltage is proportional to the rate of change of the current

$$U_L = L \frac{dI_L}{dt}$$